

Ryan Cho

Senior Full Stack Engineer | Payments, Billing & Ledger Systems

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PROFESSIONAL SUMMARY

Senior full stack engineer with 6+ years building production payment, billing, and ledger systems in fintech and capital markets environments. Deep expertise in Python, Django, and Domain-Driven Design for modular backend architecture. Proven track record owning reconciliation workflows, Stripe subscription billing, and fraud signal integrations while delivering full stack features across ops-critical financial products.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Owned **end-to-end ledger reconciliation** pipelines syncing third-party processor invoices against internal ledgers, reducing manual ops review cycles by **40%** across multi-source payment data.
- Architected **Django-based subscription billing** modules using **Domain-Driven Design** boundaries, enabling clean separation of payment, entitlement, and risk domains across a **modular production codebase**.
- Built **Stripe-integrated payment workflows** supporting recurring charges, proration, and webhook-driven state sync for **SaaS billing operations** serving thousands of active accounts.
- Implemented **per-user lifetime value and risk metric tracking** within the ledger system, giving product and finance teams **real-time visibility** into cohort performance and exposure.
- Led **full stack delivery** of deposit attribution features including **dashboard views** and client-facing UI, rolling out safely behind **feature flags** with staged production validation.
- Developed a **fraud signal integration** rules engine processing behavioral triggers such as burst account activity and auto-debit patterns, tightening **risk controls** without degrading user conversion.
- Partnered with card operations to support **physical and virtual card delivery** workflows, coordinating backend fulfillment logic ahead of **time-bound product milestones**.
- Drove **SFTP-based data ingestion** flows for external financial feeds, ensuring reliable reconciliation handoffs between vendor platforms and **internal ledger services**.
- Mentored engineers on **Python backend patterns** and cross-team architectural standards, producing **handoff documentation** that accelerated onboarding during team transitions.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Engineered **Python services** supporting trade settlement and **ledger reconciliation** across internal banking systems, improving daily close accuracy for **high-volume transaction workflows**.
- Built **payment processing integrations** connecting core banking platforms with external vendors, applying **DDD-style module boundaries** to keep reconciliation logic isolated and testable.
- Automated **invoice-to-ledger matching** routines that reduced exception queues by **30%**, enabling ops teams to focus on true discrepancies rather than routine sync failures.
- Developed **RESTful APIs and internal dashboards** for finance stakeholders to monitor subscription-style billing anomalies and **payment lifecycle states** in production.
- Collaborated with risk and compliance teams to implement **audit-ready reconciliation reports**, supporting regulatory review of **multi-currency payment flows**.
- Optimized **batch reconciliation jobs** processing millions of records nightly, cutting runtime by **25%** through query tuning and incremental sync strategies.
- Contributed to **cross-functional architecture reviews** for modular backend services, aligning new payment features with existing **domain-driven service layouts**.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **React and JavaScript** client interfaces for restaurant inventory management, improving data entry speed for **on-site staff workflows**.
- Implemented **REST API integrations** between front-end components and backend services, ensuring reliable sync of **transaction and billing records**.
- Wrote **Python utility scripts** to automate reporting pipelines, reducing manual data preparation time for **weekly operational summaries**.
- Participated in **code reviews and agile ceremonies**, delivering feature increments aligned with product milestones across a **distributed remote team**.
- Resolved **production defects** in payment display modules, restoring accurate charge visibility for **customer-facing checkout flows**.
- Documented **API contracts and deployment steps** to support knowledge transfer as the engineering team scaled.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web pages** using HTML, CSS, and JavaScript for a nonprofit foundation site, improving **mobile accessibility** across key donor touchpoints.
- Integrated **third-party donation forms** with backend tracking, enabling staff to reconcile **contribution records** against campaign goals.
- Maintained **content management workflows** and fixed cross-browser rendering issues, keeping public-facing pages **stable during fundraising cycles**.
- Collaborated with stakeholders to translate **non-technical requirements** into iterative web releases delivered on tight timelines.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, JavaScript, TypeScript, HTML/CSS, Dashboard UI, Feature-flagged rollouts, Responsive web development

Backend: Python, Django, Domain-Driven Design, REST APIs, Stripe integrations, Subscription billing, Ledger reconciliation, SFTP data flows, Webhook processing

Databases: PostgreSQL, MySQL, Redis, SQL query optimization, Transactional data modeling

Cloud & DevOps: AWS, Docker, CI/CD pipelines, Feature flags, Infrastructure automation, Secure file transfer workflows

Monitoring & Tools: Sentry, Datadog, Git, Jira, Stripe Dashboard, Postman, AI-assisted development tooling

Workflow & PM: Agile/Scrum, Cross-functional stakeholder collaboration, Technical documentation, Mentorship, Ops-critical workflow ownership